

TEST PLAN

Team Members

1. Abigail Johnson Inyang
2. Chinelo Nnamdi Kanu
3. Providence Inyange
4. Moses Kolaan

Selected Project: Bus Route & Ticketing Management System

Class Being Tested: Route

Test ID	Method Being Tested	Scenario Description	Input(s)	Expected Output / Behaviour
R-TC1	__init__() / route creation	Create a valid route	origin='Route 66', destination='District 9', distance=12.5, fare=30	Route stores the supplied values correctly.
R-TC2	__str__()	Display saved route details	Valid Route object	Readable route details are displayed.
R-TC3	__init__() / storage	Store a decimal distance	distance_km=12.5	Decimal distance is stored correctly.
R-TC4	__init__() / storage	Store the route base fare	base_fare=30	Base fare is stored correctly.
R-TC5	__init__() / storage	Use route text containing letters and numbers	origin='Route 66', destination='District 9'	Route text is stored because it contains alphabetic characters.

Class Being Tested: RouteManager

Test ID	Method Being Tested	Scenario Description	Input(s)	Expected Output / Behaviour
RM-TC1	add_route()	Add a valid route	Route 66, District 9, 12.5, 30	Route is created and stored.
RM-TC2	add_route()	Origin is empty	origin=""	Route is rejected with a clear message.
RM-TC3	add_route()	Destination contains only numbers	destination='123'	Route is rejected.
RM-TC4	add_route()	Distance is negative	distance_km=-1	Route is rejected.
RM-TC5	delete_route()	Delete a route linked to an existing trip	Existing route ID with trip records	Deletion is prevented.

Class Being Tested: Trip

Test ID	Method Being Tested	Scenario Description	Input(s)	Expected Output / Behaviour
TR-TC1	__init__() / trip creation	Create a valid trip	route_id=1, date=2026-12-17, 08:00-09:00, seats=4	Trip stores the supplied values correctly.
TR-TC2	__str__()	Display trip details	Valid Trip object	Readable trip details are displayed.
TR-TC3	__init__() / storage	Store the travel date	travel_date=2026-12-17	Travel date is stored correctly.
TR-TC4	__init__() / storage	Store seat capacity	total_seats=4	Seat count is stored correctly.
TR-TC5	__init__() / storage	Store valid departure and arrival times	departure='08:00', arrival='09:00'	Times are stored correctly.

Class Being Tested: TripManager

Test ID	Method Being Tested	Scenario Description	Input(s)	Expected Output / Behaviour
TM-TC1	add_trip()	Create a valid trip	Valid route, future date, valid times, 4 seats	Trip is created and stored.
TM-TC2	search_trips()	Search by existing route ID and date	route_id=1, date=2026-12-15	Matching trip is returned.
TM-TC3	update_trip()	Update trip date and times	date=2026-12-18, 10:00-11:00	Trip is updated successfully.
TM-TC4	cancel_trip()	Cancel a trip with an active ticket	Existing trip with active ticket	Cancellation is prevented.
TM-TC5	view_seat_layout()	View seats on a trip	Trip with four open seats	Seat map shows the current seat availability.

Class Being Tested: Passenger

Test ID	Method Being Tested	Scenario Description	Input(s)	Expected Output / Behaviour
P-TC1	__init__() / passenger creation	Create a valid passenger	name='Emma Audit', phone='59900112'	Passenger stores the supplied values correctly.
P-TC2	__str__()	Display passenger details	Valid Passenger object	Readable passenger details are displayed.
P-TC3	__init__() / storage	Store phone number as text	phone='59900112'	Phone number is stored correctly.
P-TC4	__init__() / storage	Store a full name containing spaces	name='Alice Audit'	Full name is stored correctly.
P-TC5	__init__() / update result	Store a valid updated passenger name	name='Alice Updated'	Updated name is stored correctly.

Class Being Tested: PassengerManager

Test ID	Method Being Tested	Scenario Description	Input(s)	Expected Output / Behaviour
PM-TC1	register_passenger()	Register a valid passenger	name='Emma Audit', phone='59900112'	Passenger is created and stored.
PM-TC2	register_passenger()	Name is empty	name=""	Registration is rejected.
PM-TC3	register_passenger()	Name contains only numbers	name='12345'	Registration is rejected.
PM-TC4	search_passenger()	Search for an existing passenger ID	passenger_id=1	Matching passenger is returned.
PM-TC5	update_passenger()	Valid new name with invalid phone	name='Changed Name', phone='WEW'	Update is rejected and the original passenger data remains unchanged.

Class Being Tested: BusPass

Test ID	Method Being Tested	Scenario Description	Input(s)	Expected Output / Behaviour
BP-TC1	__init__() / pass creation	Create a valid student pass	passenger=4, type='student', 2026-12-10 to 2026-12-20	BusPass stores the supplied values correctly.
BP-TC2	__str__()	Display bus pass details	Valid BusPass object	Readable bus pass details are displayed.
BP-TC3	__init__() / storage	Create a senior pass	pass_type='senior'	Senior pass type is stored correctly.
BP-TC4	__init__() / storage	Create a priority pass	pass_type='priority'	Priority pass type is stored correctly.
BP-TC5	status storage	Suspend a bus pass	Existing active pass	Pass status changes to suspended.

Class Being Tested: BusPassManager

Test ID	Method Being Tested	Scenario Description	Input(s)	Expected Output / Behaviour
BPM-TC1	issue_pass()	Issue a valid student pass	Existing passenger, student, valid dates	Pass is created and stored.
BPM-TC2	issue_pass()	Passenger does not exist	passenger_id=999	Pass is not created.
BPM-TC3	issue_pass()	Passenger already has an active pass	Existing passenger with active pass	Second active pass is rejected.
BPM-TC4	renew_pass()	Renew an existing pass with a later expiry date	pass_id=1, new_expiry=2027-01-01	Pass is renewed successfully.
BPM-TC5	renew_pass()	Renew an old pass while another active pass exists	Suspended old pass + newer active pass	Renewal is rejected so only one active pass remains.

Class Being Tested: Ticket

Test ID	Method Being Tested	Scenario Description	Input(s)	Expected Output / Behaviour
T-TC1	__init__() / ticket creation	Create a normal active ticket	passenger=4, trip=1, seat=1, price=40	Ticket stores the supplied values correctly.
T-TC2	__init__() / storage	Create a ticket without a bus pass	pass_id=None	Ticket is stored without a pass.
T-TC3	__init__() / storage	Create a ticket using a student pass	pass_id=1, price=20	Ticket stores the applied pass and discounted price.
T-TC4	cancel()	Cancel an active ticket	Existing active ticket	Ticket status changes to cancelled.
T-TC5	__str__()	Display a ticket	Existing ticket	Readable ticket details are displayed.

Class Being Tested: TicketManager

Test ID	Method Being Tested	Scenario Description	Input(s)	Expected Output / Behaviour
TKM-TC1	purchase_ticket()	Buy a normal ticket without a pass	Existing passenger/trip, open seat, no pass	Ticket is created at the full route fare.
TKM-TC2	purchase_ticket()	Buy a ticket with a valid student pass	Valid student pass	Ticket is created at 50% of the route fare.
TKM-TC3	purchase_ticket()	Choose an occupied seat	Seat already used by an active ticket	Purchase is rejected.
TKM-TC4	purchase_ticket()	Choose a seat above trip capacity	seat=5 on a 4-seat trip	Purchase is rejected.
TKM-TC5	purchase_ticket()	Use an expired pass	Expired student pass	Ticket is created at full fare without the pass benefit.